Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 2

Upload and analyze a CSV dataset and perform preprocessing and visualization

Experiment 2: Upload and analyze a CSV dataset and perform preprocessing and visualization

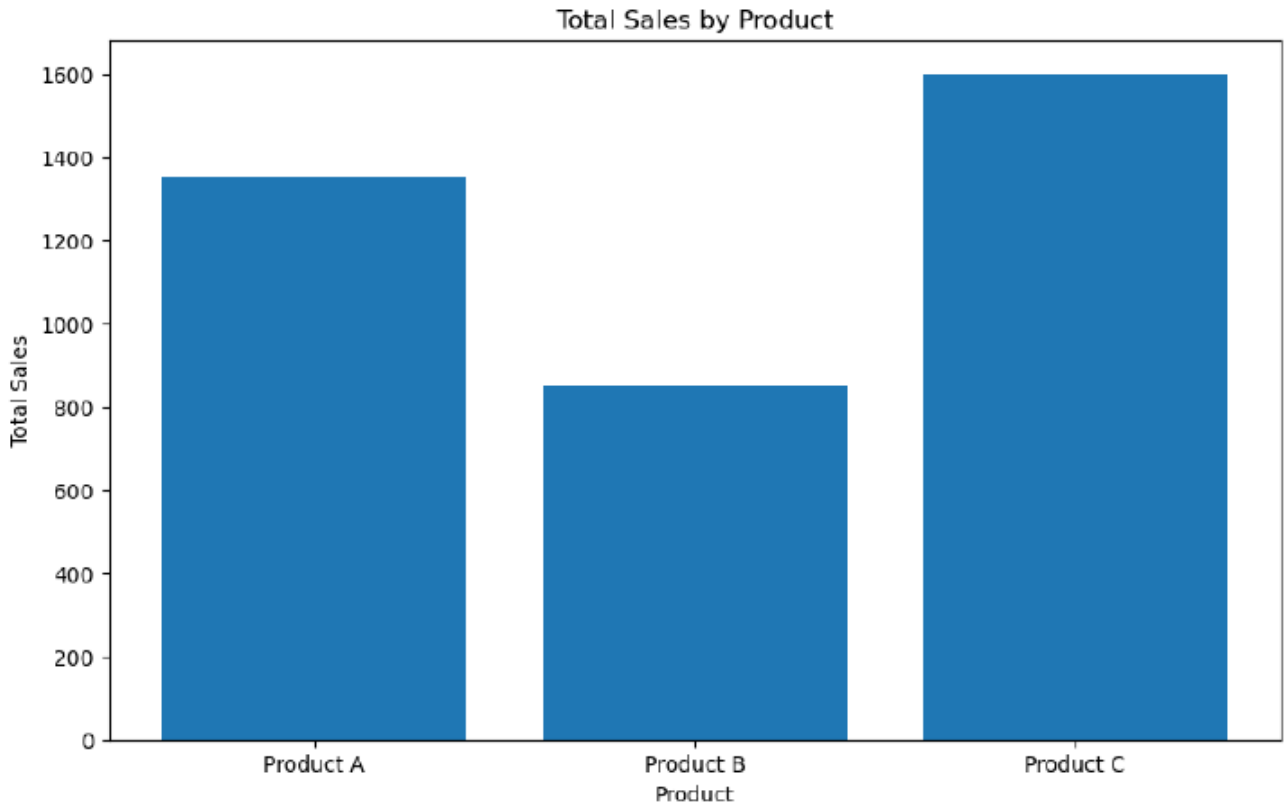
Aim

To analyze a CSV dataset by performing preprocessing and visualization.

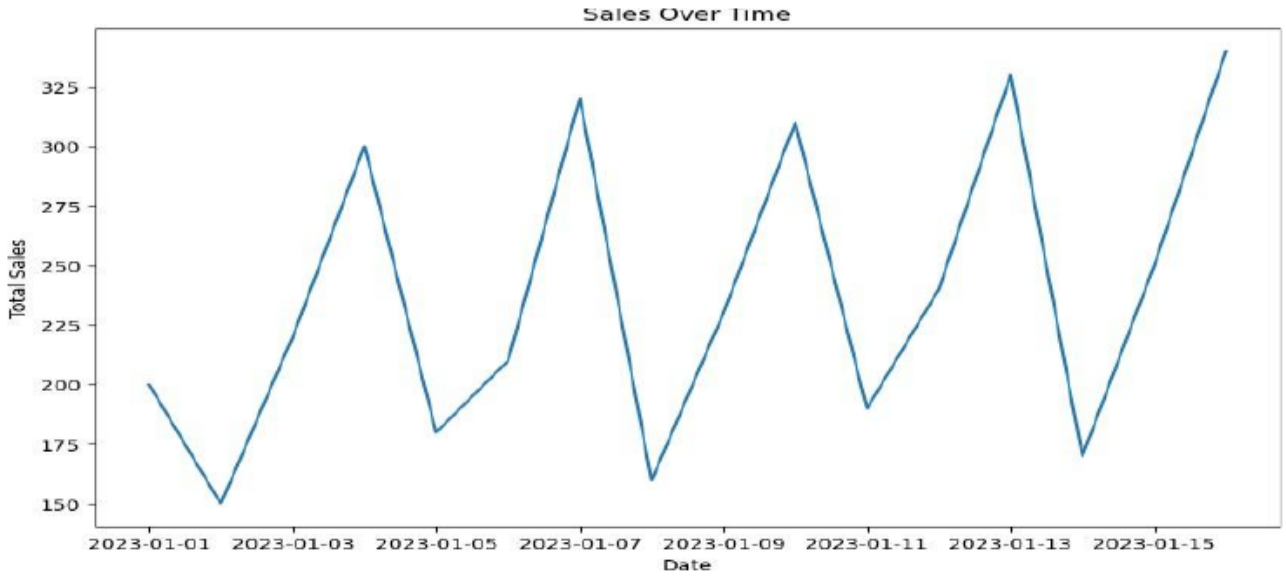
Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
print(df.head())
print(df.isnull().sum())
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
correlation_matrix = df.corr()
print(correlation_matrix)
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

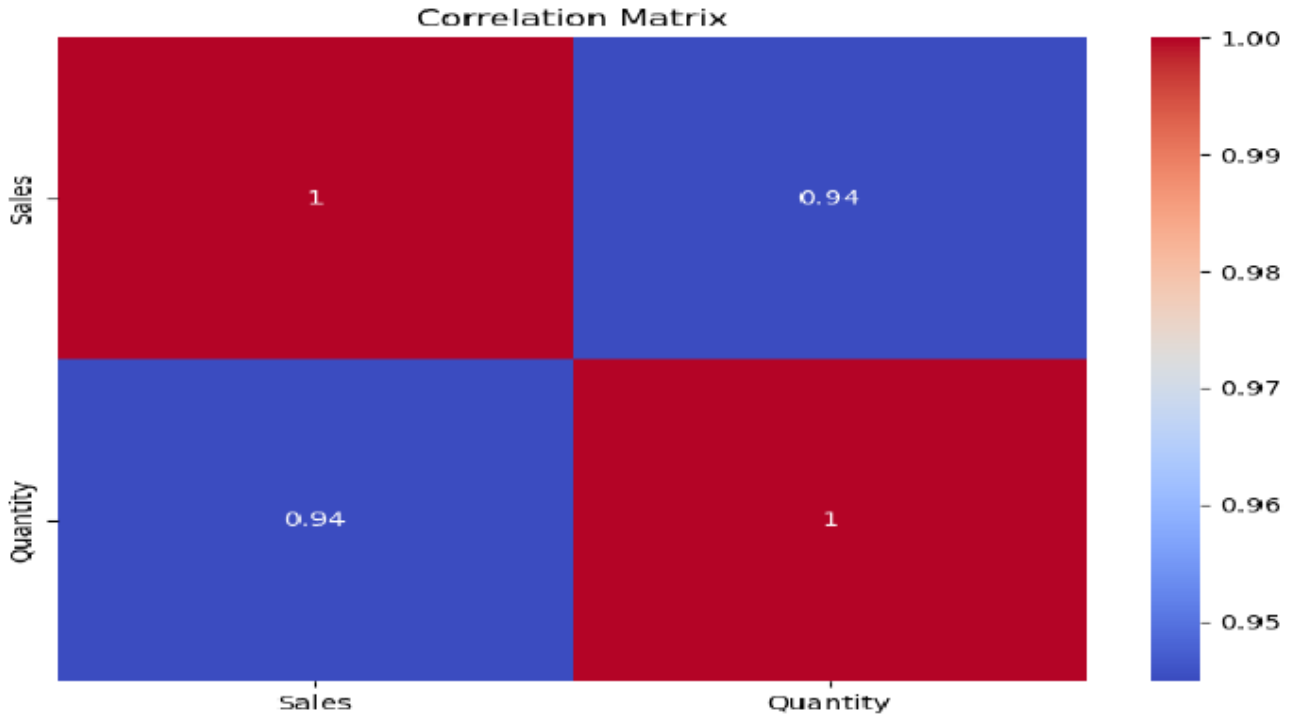
Output - Experiment 2



Output - Experiment 2



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Output - Experiment 2

Product Product A Product B Product C
Region
East 0 0 1600
North 1350 0 0
South 0 480 0
West 0 370 0
Sales Quantity
Sales 1.000000 0.944922
Quantity 0.944922 1.000000

Result - Experiment 2

The CSV dataset was analyzed, preprocessed and visualized.